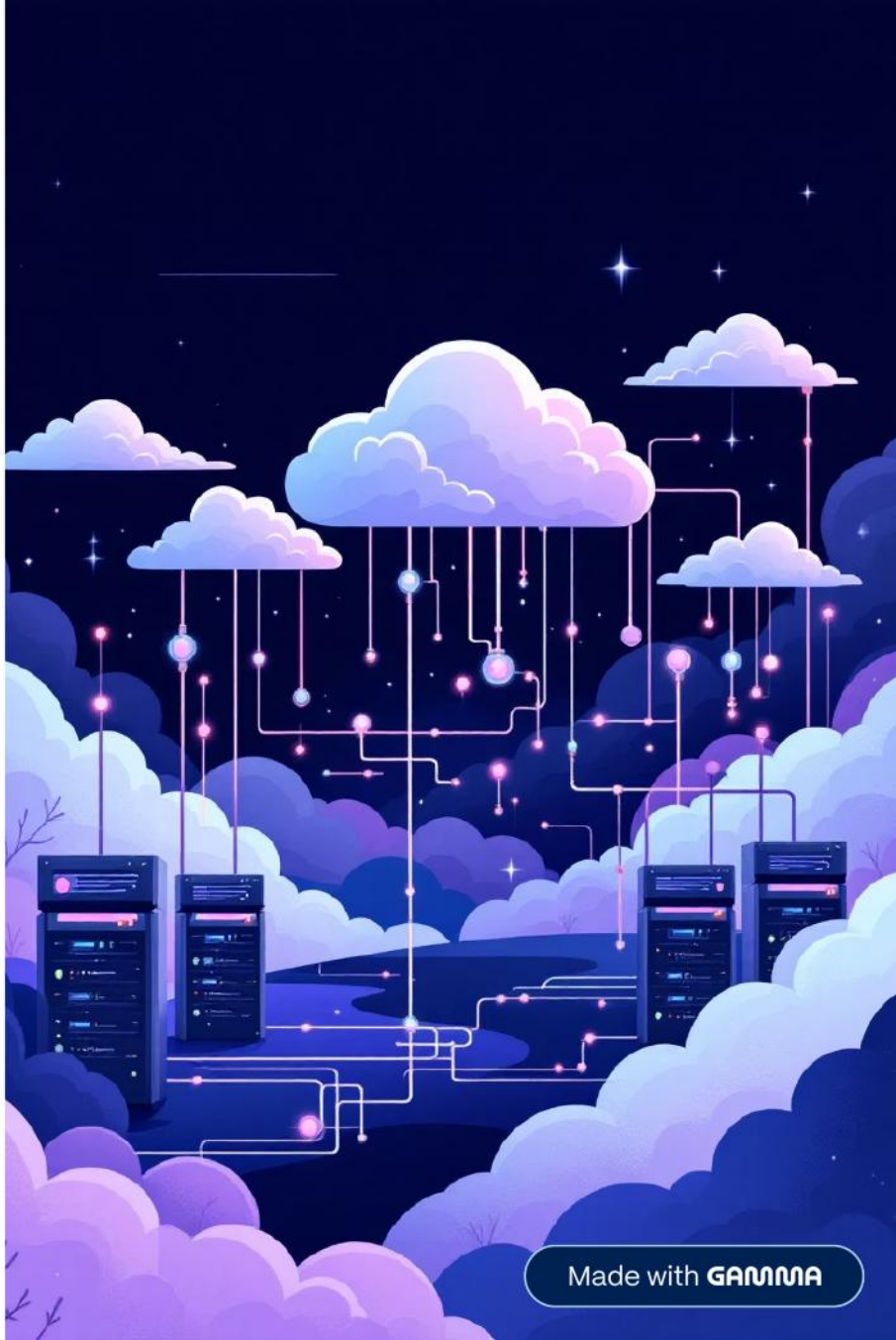
AWS Application Load Balancer with EC2 – High Availability Architecture

A production-ready architecture demonstrating traffic distribution and fault tolerance using AWS services



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Project Overview

What This Project Demonstrates

This project showcases how an AWS Application Load Balancer (ALB) intelligently distributes incoming web traffic across multiple EC2 instances to achieve high availability and fault tolerance.

Rather than depending on a single server, the application operates across multiple EC2 instances, with the load balancer routing user requests exclusively to healthy instances.

This architecture represents a real-world production setup commonly deployed in enterprise cloud-based applications.

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Why Application Load Balancer?

The Problem

Single EC2 deployments create critical vulnerabilities:

- Single point of failure
- No fault tolerance
- Limited scalability

The Solution

Application Load Balancer addresses these challenges by:

- Distributing traffic intelligently
- Handling instance failures automatically
- Ensuring continuous availability

The ALB functions as a unified entry point whilst efficiently managing backend server resources.

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Architecture Advantages

High Availability

Application remains accessible even when individual EC2 instances experience failures or maintenance downtime.

Fault Tolerance

Traffic routes exclusively to healthy instances based on continuous health check monitoring.

Production-Ready Design

Architecture matches real industry standards and easily integrates with Auto Scaling for future growth.

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AWS Services Used

1

Amazon EC2

Virtual compute service hosting the web application instances across multiple availability zones.

2

Application Load Balancer

Distributes incoming HTTP traffic across EC2 instances based on health checks and routing rules.

3

Target Groups

Logical grouping of EC2 instances that receive and handle traffic forwarded from the load balancer.

4

Virtual Private Cloud

Provides isolated networking environment with subnets, route tables, and internet gateway configuration.

5

Security Groups

Acts as virtual firewall controlling inbound and outbound traffic to EC2 instances and ALB.

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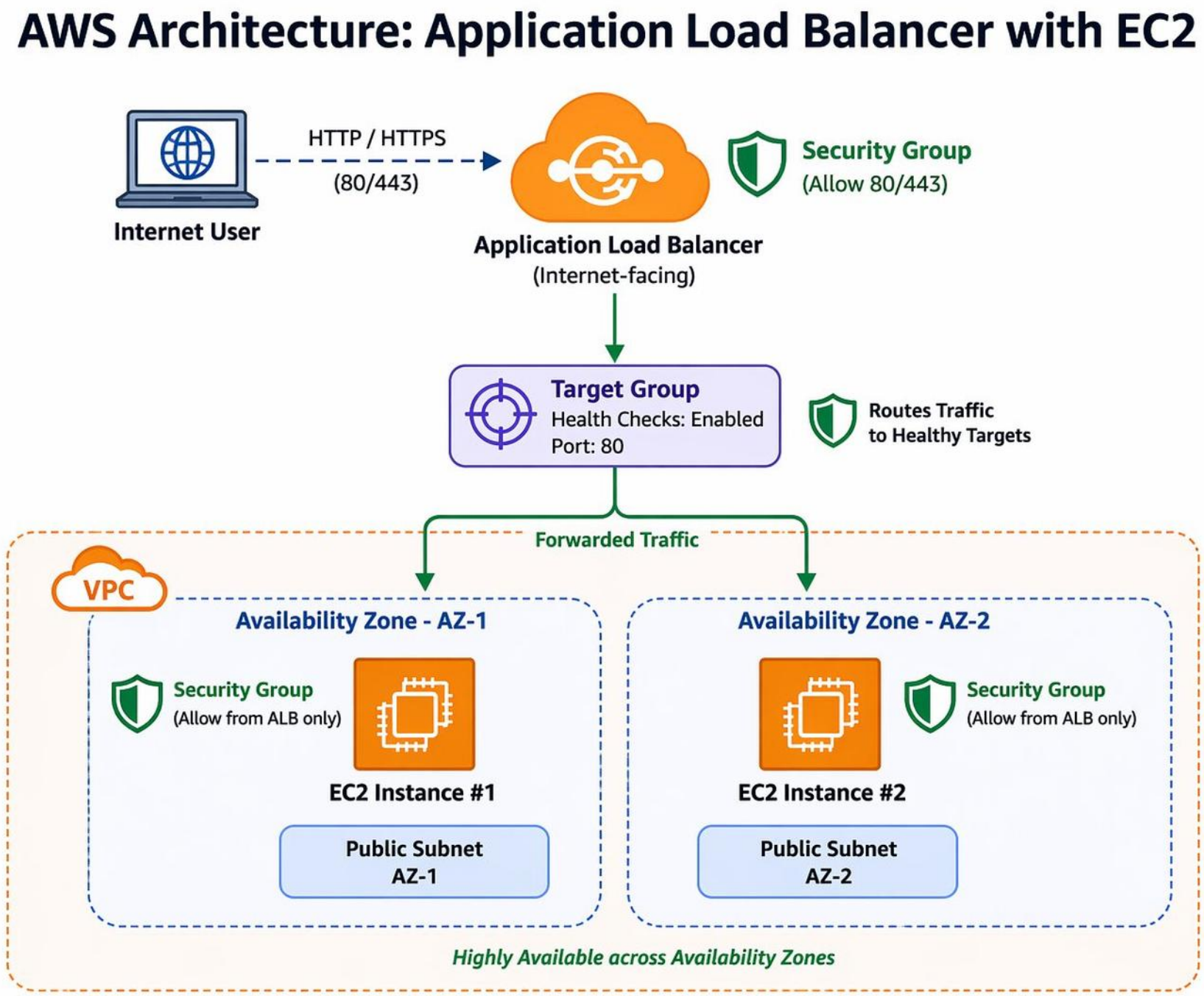
Architecture Diagram



This architecture shows how an Application Load Balancer distributes incoming traffic across multiple EC2 instances deployed in different Availability Zones. The design ensures high availability, fault tolerance, and scalable web application hosting on AWS.

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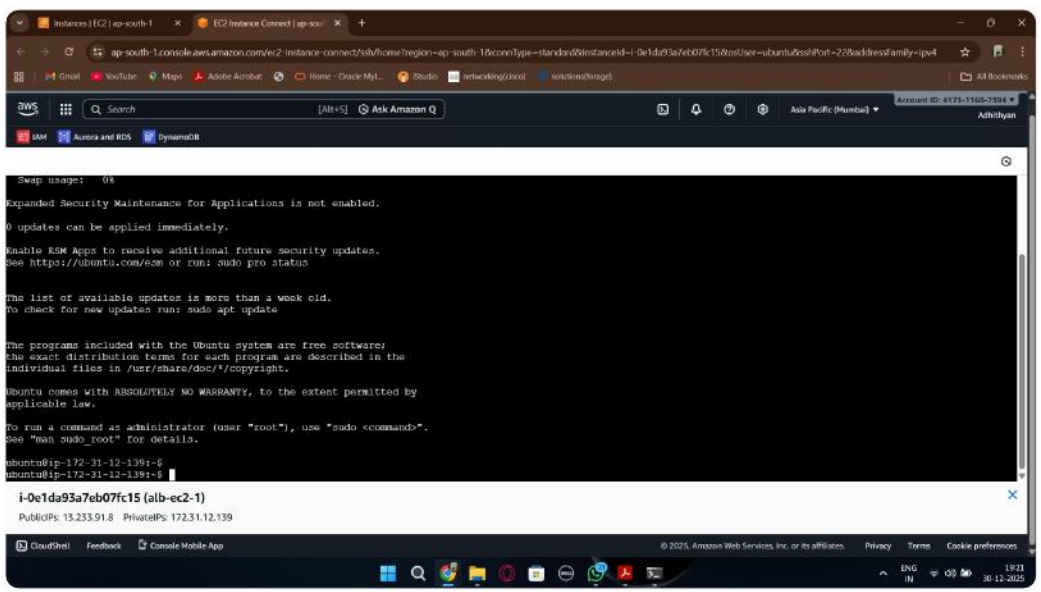
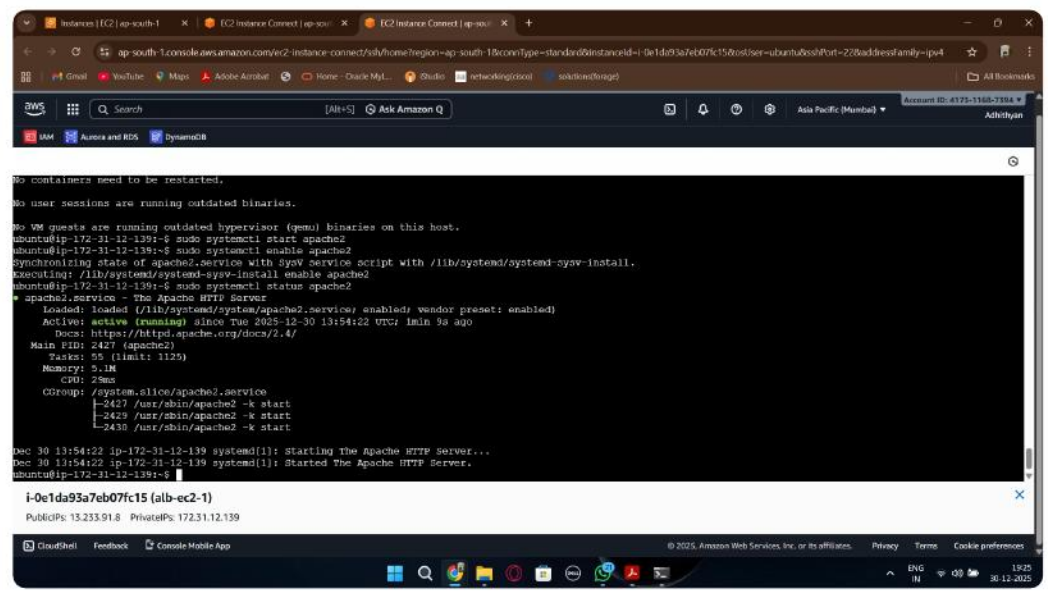
EC2 Instance Connect & Apache Installation

EC2 Instance Connect (Terminal Access)



This screenshot shows successful connection to the EC2 instance using EC2 Instance Connect. It confirms that the instance is running and accessible for server configuration.

Apache Service Installation & Status



Apache web server is installed and started on the EC2 instance using systemctl commands. The service status shows Apache is active and running successfully.

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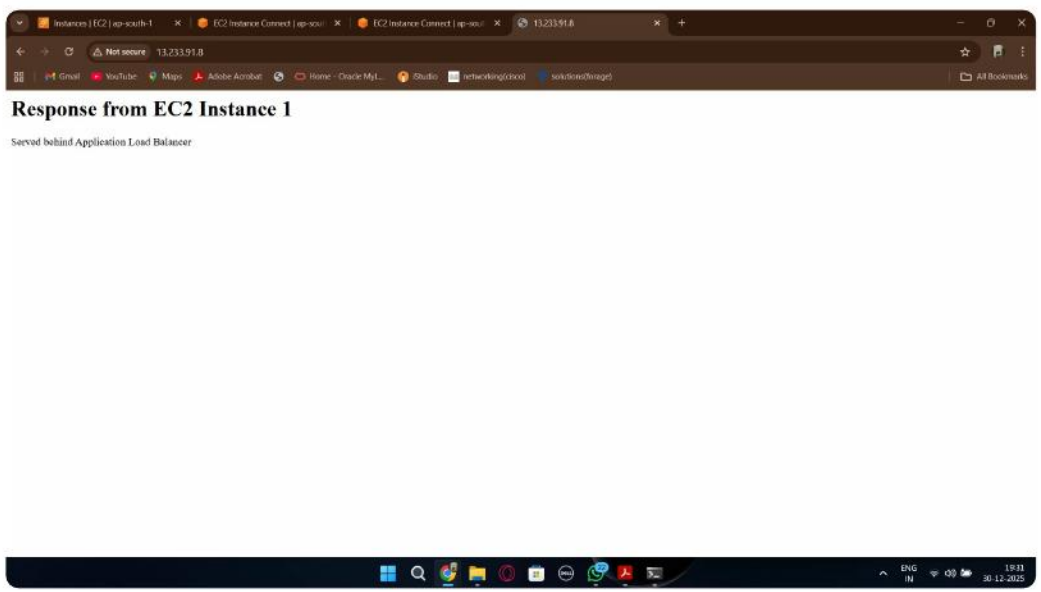
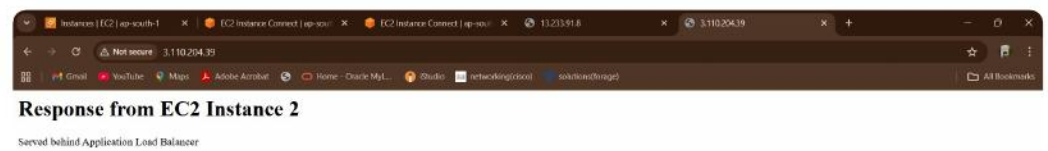
Web Page Response from EC2 Instances

Web Page Response from EC2 Instance 1



This output confirms that EC2 Instance 1 is serving HTTP requests correctly. The custom message helps identify responses coming from this specific instance.

Web Page Response from EC2 Instance 2



This output verifies that EC2 Instance 2 is also serving HTTP traffic properly. Different response text is used to distinguish traffic routing between instances.

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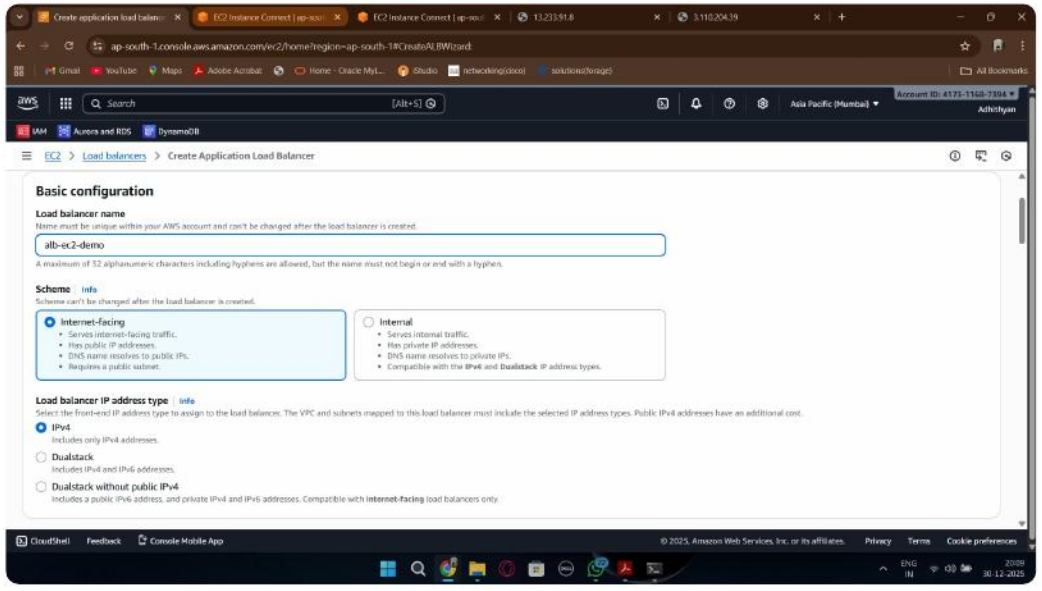
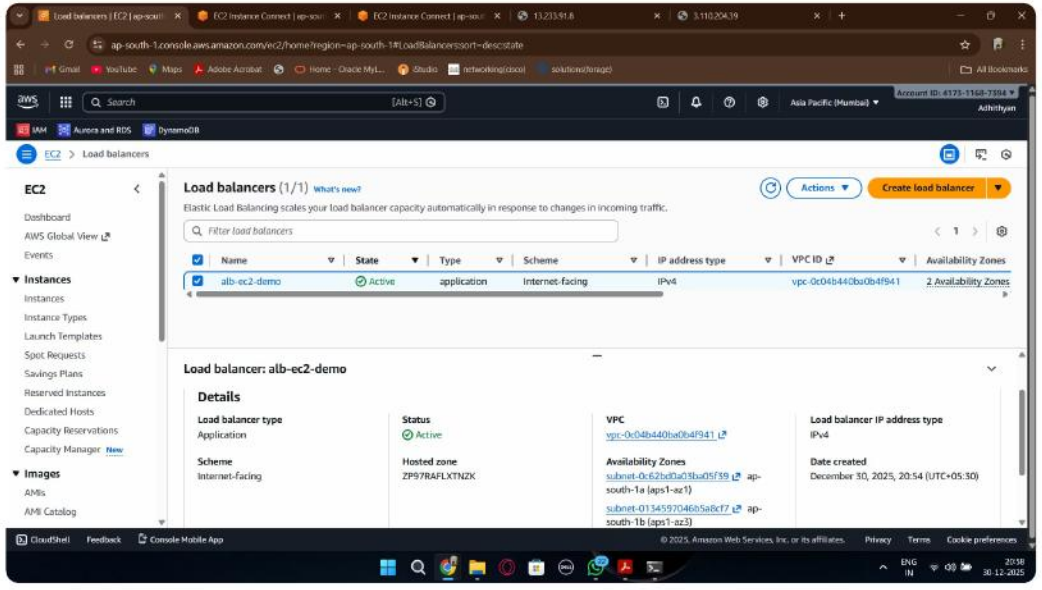
Application Load Balancer Creation (Basic Configuration)

This screen shows the creation of an Internet-facing Application Load Balancer. The ALB is configured to distribute incoming HTTP traffic across multiple AZs.



Load Balancer Active State

The Application Load Balancer is successfully created and in an active state. It confirms that the ALB is ready to accept and route client requests.



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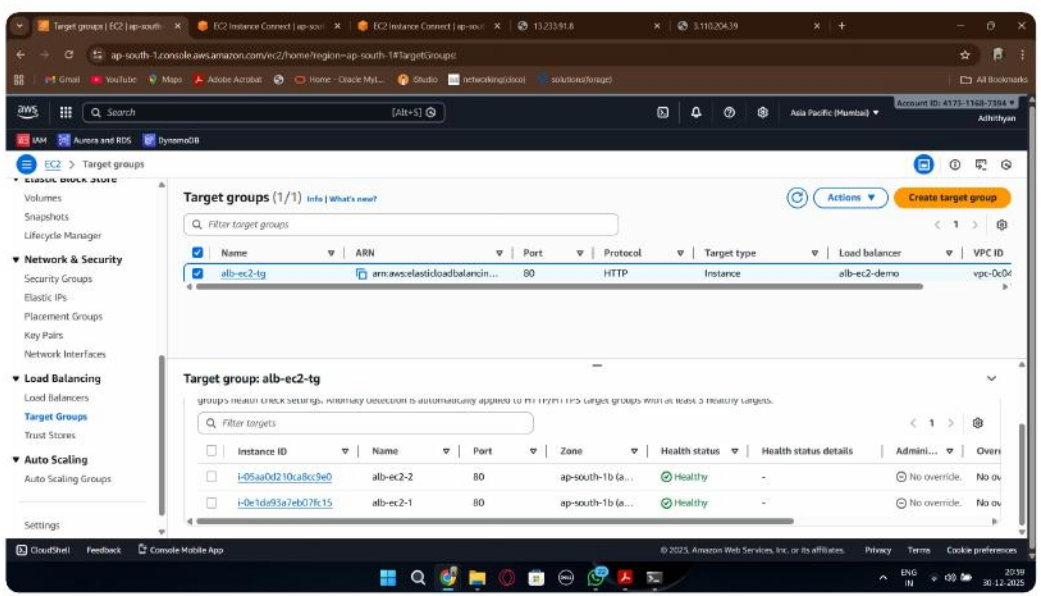
Target Group & Load Balancer Configuration

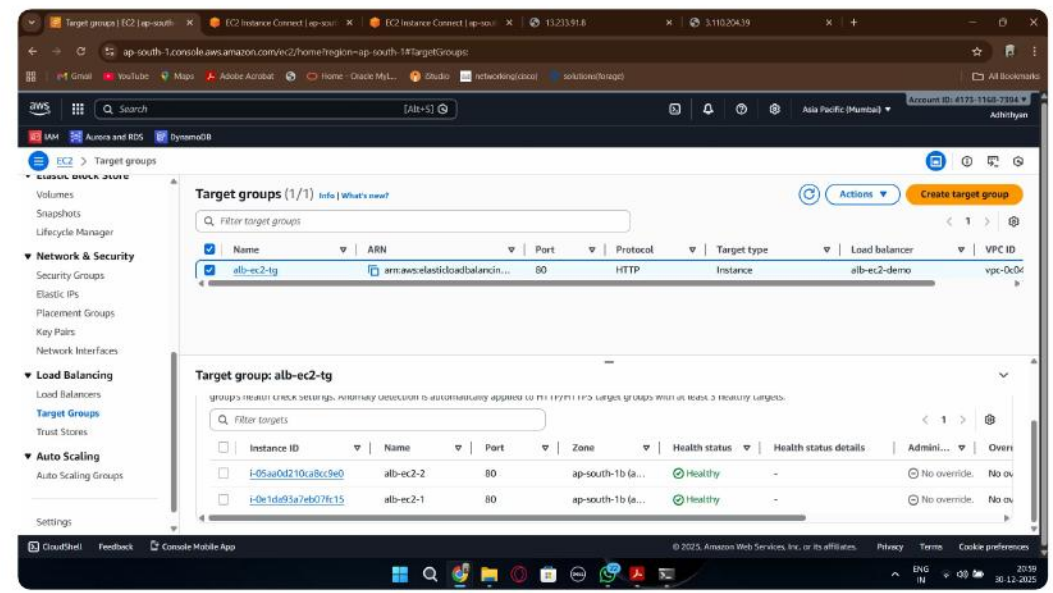
Target Group Attached to Load Balancer

This screenshot shows the target group linked to the Application Load Balancer. Traffic received by the ALB will now be forwarded to healthy EC2 instances.



Target Group Details

Complete target group configuration showing health check settings and instance registration. This ensures only healthy instances receive traffic from the load balancer.



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Key Takeaways

This project demonstrates the deployment of a highly available web application using AWS Application Load Balancer and EC2 instances. By distributing traffic across multiple EC2 instances in different Availability Zones, the solution ensures improved reliability, scalability, and fault tolerance.

Hands-on AWS Experience

Practical implementation of EC2, ALB, Target Groups, Security Groups, and VPC networking concepts.

Production-Ready Architecture

Real-world cloud architecture practices reflecting industry standards for resilient applications.

Foundation for Scaling

Solid groundwork for integrating Auto Scaling groups and advanced load balancing strategies.

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